

ECON211: Quiz 1B

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Instructions: general

- DO NOT TURN THIS PAGE BEFORE THE EXAM STARTS!
- Write your full name above.
- You cannot use notes, books, or electronic equipment including phones, headphones, calculators, etc.
- Use scratch paper or the back of the exam pages for your computations.
- No communication or gazing into others' papers.
- You have **50 minutes**.
- There are no partial points.
- After time runs out, you must close the exam and put it in front of you **immediately**.
- If you violate any of the above rules, you fail the exam.
- Good luck!

Instructions: how to write your answers

- Your final answer must be either a number, an arithmetic or an algebraic expression, or a short sentence, depending on the question. You can leave your answer in terms of fractions, binomials, exponents, etc., that is no need to compute an exact numerical value, an arithmetic expression is sufficient.
- Your solutions must be written at the last page of the exam or on a separate signed page. I will consult your solutions in case of an ambiguity.
- Circle each of your final answers! Points are awarded for correct final answers; solutions are only consulted for clarification.
- You cannot use pencils to write down your answers.

	$X = 0$	$X = 1$	$X = 2$
$Y = 1$	0	0.25	0.2
$Y = 2$	0.2	0.25	0.1

Joint distributions

There are two assets. Let X and Y be random variables equal to the return on the first asset and the second asset, respectively. The joint distribution of X and Y is given by the table above.

1. (3 points) Find the probability that Y is equal to 1.
2. (3 points) Suppose you know that $Y = 1$. Find the probability that $X = 1$.
3. (4 points) Find the probability that $X \geq Y$.
4. (4 points) Find the expected value of Y .
5. (4 points) Find the variance of Y .
6. (4 points) Find $E(Y|X = 2)$.
7. (5 points) Find the covariance between X and Y .
8. Let $\alpha \in [0, 1]$ be the share of the first asset in a portfolio.
 - (a) (3 points) Find α that maximizes the expected return of the portfolio.
 - (b) (5 points) Find α that minimizes the variance of return of the portfolio.

First price sealed bid auction

Alice is selling her car using a first price sealed bid auction. There are 2 potential buyers. Buyers have valuations that are independently and identically distributed with the following cumulative distribution function (CDF):

$$F(x) = \begin{cases} x^2 & \text{if } x \in [0, 1], \\ 0 & \text{otherwise.} \end{cases}$$

Buyers know their own valuations, while Alice does not. Each buyer i submits a bid b_i . Suppose each buyer bids $b_i = \frac{v_i}{2}$. Alice sells the car to the highest bidder, who pays the amount of their own bid.

1. (4 points) Let X be the random variable denoting Alice's revenue. Compute $\Pr(X \leq 1/3)$.
2. (5 points) Compute $E[X]$.

Write your solutions here.